

Unit 2 - Proofs and Logic

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The following B.E.S.T. Standards will be covered in this unit:

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

- *Clarification 1: Postulates, relationships, and theorems including vertical angles are congruent; when a transversal crosses parallel lines, the consecutive angles are supplementary and alternate (interior and exterior) angles and corresponding angles are congruent; points on a perpendicular bisector of a line segment are exactly those equidistant from the segment's endpoints.*
- *Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flowchart proofs or informal proofs.*
- *Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.*

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.

- *Clarification 1: Within the Geometry course, instruction focuses on the connection to proofs within the course.*

MA.912.LT.4.3 Identify and accurately interpret “if...then,” “if and only if,” “all” and “not” statements. Find the converse, inverse and contrapositive of a statement.

- *Clarification 1: Instruction focuses on recognizing the relationships between an “if...then” statement and the converse, inverse and contrapositive of that statement.*
- *Clarification 2: Within the Geometry course, instruction focuses on the connection to proofs within the course.*

Unit 2, Lesson 1: Introduction to Euclidean Geometry



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can distinguish between definition, postulate, and theorem.
- ☐ I understand the role of proof.
- ☐ I can relate properties of equality to the properties of congruence.



2.1.1 Warm-Up: Number Cubes Math

A picture of 16 number cubes is shown.



1. What do you notice?
2. Find the sum of the number cubes. Show your work or explain your thinking.
3. If all of the number cubes with 3 dots were changed to number cubes with 6 dots, how would you find the new sum? Show your work or explain your thinking.



2.1.2 Exploration: What is Euclidean Geometry?

Euclidean geometry is the study of planes and solid figures on the basis of axioms and theorems compiled by the Greek mathematician Euclid. Rather than using algorithms to solve problems like you may have done in other math courses, Euclidean geometry emphasizes using facts, ideas, and assumptions.



Euclid is known as the father of Geometry. He wrote 13 books using an axiomatic system to construct bodies of knowledge.

Each of Euclid's 13 books have proofs. Each proof shows how a hypothesis is true. The proof uses definitions, postulates, and proven theorems to explain why the hypothesis is true.

A **postulate** is an idea that is assumed to be true or something that is self-evident.

The first postulate in Euclid's Book 1 is shown.

A straight line segment can be drawn joining any two points.

1. Use a straightedge to show the postulate using the points shown.

R



• W

2. Discuss with your partner why a postulate is something that is self-evident. Summarize your discussion.

A **theorem** is an idea that has to be shown to be true through a chain of reasoning, often called a proof.

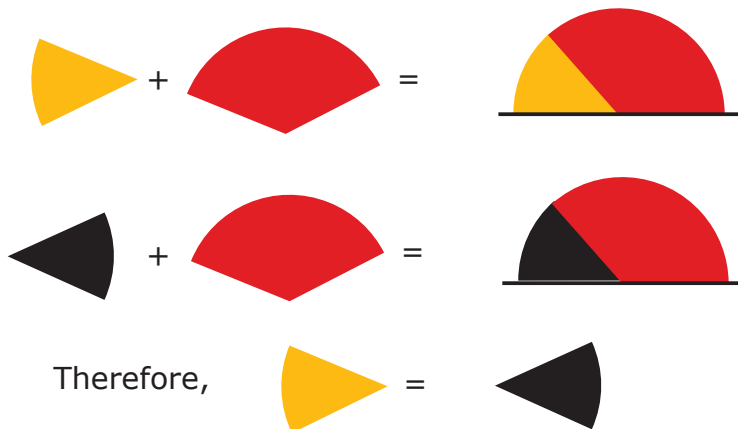


Vertical Angles Theorem: If two lines intersect, then vertical angles are congruent.

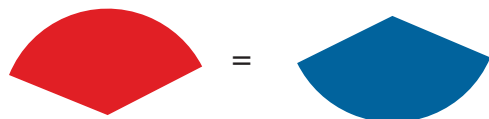
The vertical angles are



A pictorial proof of the vertical angles theorem is shown.



In the same manner,



3. Discuss with your partner the pictorial proof of the vertical angles theorem below. Complete the description for each picture shown in the pictorial proof.

Picture	Description
	The yellow angle and the red angle are being added.

Proofs play an important role in the course of geometry. Proofs can take on many forms. They are the set of logical steps that explain why something is true.

When programmers write a new game, they use a set of logical steps. When you tie your shoes, you use a set of logical steps.

4. Work with your partner to place the steps in tying a pair of shoes in order.

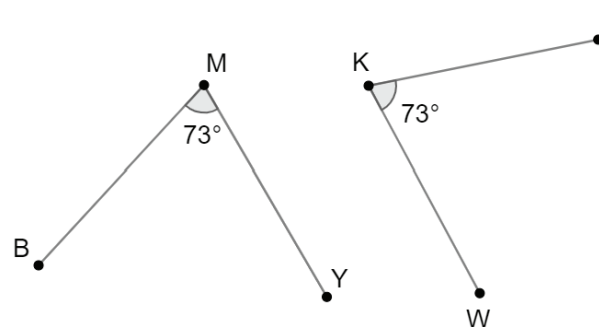
Step	Description	Picture
	Make a loop in one lace.	
	Cross the laces and pull them tight.	
	Pull both loops tight.	
	Wrap the other lace around the loop and push some lace through the hole that is made at the bottom to make a second loop.	

5. Discuss with your partner why someone who learned to tie their shoes using a different method may have difficulty putting these steps in a logical order. Summarize your discussion.



2.1.3 Guided Instruction: Properties of Equality and Congruence

In geometry, there are different notations for congruence and equality. For example, $\angle BMY$ and $\angle LKW$ have the same measure, so the angle measurements are equal. This is written as $m\angle BMY = m\angle LKW$. It can also be said that the angles are congruent, which is written as $\angle BMY \cong \angle LKW$.



Equality and congruence may seem to be the same, but equality is used when relationships can be expressed using numbers. Congruence is used when the relationship is not noted using numbers.

1. For each property, complete the example(s).

a. Reflexive property

Equality	Congruence
$m\angle WHD = m\angle \underline{\hspace{2cm}}$	$\angle WHD \cong \angle \underline{\hspace{2cm}}$

b. Symmetric property

Equality	Congruence
If $m\angle CKR = 59^\circ$ then $\underline{\hspace{2cm}}$.	If $\angle CKR \cong \angle HDE$, then $\underline{\hspace{2cm}}$.

c. Transitive property

Equality	Congruence
If $m\angle CRK = m\angle HWD$ and $m\angle HWD = 90^\circ$, then $m\angle CRK = \underline{\hspace{2cm}}$.	If $\angle CRK \cong \angle HWD$ and $\angle HWD \cong \angle MPX$, then $\angle CRK \cong \underline{\hspace{2cm}}$.

d. Substitution property

Equality
If $m\angle CRK + m\angle HWD = 156^\circ$ and $m\angle CRK = 84^\circ$, then $\underline{\hspace{2cm}} + m\angle HWD = 156^\circ$.

2. Write the conclusion for each statement. Then, name the property.

Statement	Property
If $m\angle FDT + m\angle BYH = 245^\circ$ and $245^\circ = m\angle MPW$, then	
If $m\angle FDT + m\angle BYH = m\angle RLC + m\angle KNW$ and $m\angle RLC = m\angle BYH$, then	

3. Discuss with your partner the conditional statement, "If $m\angle CRK = m\angle HWD$, then $\angle CRK \cong \angle HWD$." Summarize your discussion.

In geometry, there is a formal difference between equality and congruence. Equality refers to measurements, which is why $m\angle CRK = m\angle HWD$ includes the m for measurement. On the other hand, congruence means the shape is the same, which can be shown using isometry. In other words, the figure coincides exactly when superimposed.



If two figures have equal measurements, then by the **definition of congruence** the figures are congruent. If congruence is shown through a series of transformations, then the **definition of congruence in terms of rigid motions** was applied.

The definition of congruence can be used to change from congruence to equality.

Statement	Reason
1. $\angle CRK \cong \angle HWD$	1. Given
2. $m\angle CRK = m\angle HWD$	2. Definition of congruence

Or, the definition of congruence can be used to change from equality to congruence.

Statement	Reason
1. $m\angle CRK = m\angle HWD$	1. Given
2. $\angle CRK \cong \angle HWD$	2. Definition of congruence



Gottfried Wilhelm Leibniz (1646-1716) introduced a symbol using one horizontal bar under the tilde for congruence in an unpublished manuscript of 1679. In 1824, Carl Brandan Mollweide (1774-1825) used the modern congruent symbol with two horizontal bars under a tilde in the book, *Euklid's Elemente*.



2.1.4 Wrap-Up: Cool Down

1. Complete the table.

The terms and information used in the Exploration and Guided Instruction that I knew were...	Something I wondered during the Exploration or Guided Instruction was...	Something I learned during the Exploration or Guided Instruction was...

Unit 2, Lesson 2: Types of Reasoning



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can write a conjecture.
- ☐ I can write a counterexample.
- ☐ I can use inductive and deductive reasoning.



2.2.1 Warm-Up: What's the Pattern?

Four steps of a pattern are shown.

Step 1	Step 2
Step 3	Step 4

- How many blue circles will be in step 5?
- How many orange circles will be in step 5?
- How many green circles will be in step 5?
- Describe the pattern using words or an algebraic expression.



2.2.2 Exploration: Types of Reasoning

To describe a pattern such as the pattern in the Warm-Up is to make a conjecture.

A **conjecture** is a mathematical statement that has not yet been proven.

Once a conjecture has been made, then reasoning can be used to prove the conjecture.

1. Work with your partner to write conjectures for the pattern in the Warm-Up. You should be able to use each conjecture to determine any step in the pattern.

- number of blue circles
- number of orange circles
- number of green circles
- total number of circles

Blue	Orange	Green	Total

A conjecture is an important step in problem-solving. Conjectures arise when you notice a pattern and that pattern holds true for many cases. However, just because a pattern holds true for many cases does not mean that the pattern will hold true for **all** cases. Conjectures must be proven for the mathematical observation to be fully accepted. When a conjecture is rigorously proven, it becomes a theorem.

The values for step 10 are shown. Use these values to test your conjecture. Revise your conjecture if necessary.

Blue	Orange	Green	Total
63	81	40	184

2. Ask a classmate for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the conjecture and summary columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Color	Conjecture	My Summary of Their Reason	Initials
Blue			
Orange			
Green			
Total			

The conjectures from your classmate may be different from your conjectures, but that does not mean that one is wrong. Both may be correct.

3. Compare your conjecture with the conjectures you and your partner collected from different classmates.

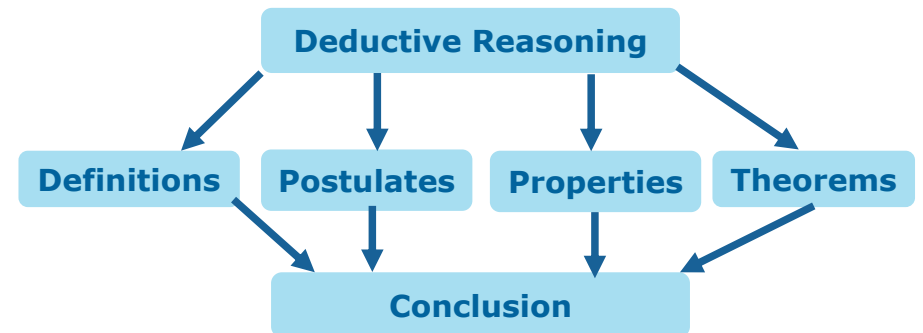
4. The values for step 18 are shown. Use these values to test your classmates' conjectures.

Blue	Orange	Green	Total
111	289	72	472

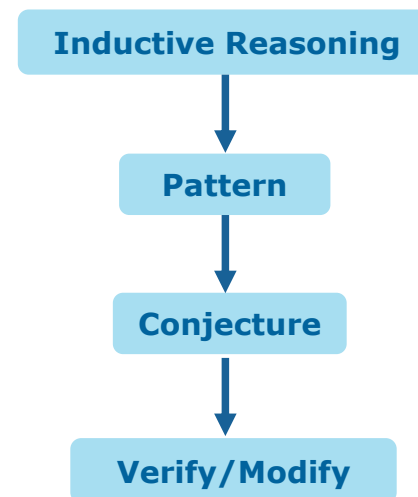
There are two types of reasoning that can be used to prove a conjecture.



Deductive reasoning uses facts and general principles to draw conclusions.



Inductive reasoning uses specific observations and patterns to draw conclusions.



5. Discuss with your partner which type of reasoning best describes the type that was used in the warm-up. Summarize your discussion.

Much of the reasoning that will be used in geometry is deductive reasoning.

Deductive reasoning needs to be used to prove the following statement is true.

All numbers ending in 0 or 5 are divisible by 5.

To verify that **all** numbers that end in 0 or 5 are divisible by 5 would require more than the method of verifying and modifying conjectures. When trying to show that something works for all cases, deductive reasoning should be used.

6. The following statement is an example of inductive reasoning.

*All small dogs in the park today have blue collars.
Therefore, all small dogs have blue collars.*

Discuss with your partner if it is possible that there existed a scenario that could have led to the conclusion. Write a short explanation.

A **counterexample** is a statement that disproves a conjecture.

7. Write a counterexample that would disprove “all small dogs have a blue collar.”



2.2.3 Bring It Together: Using Reasoning

1. Determine the type of reasoning for each statement.

Statement	Type of Reasoning
The car's battery provides power to the engine, so if the battery is dead, the car won't start.	
I know Joe is a good cook because I've eaten at his house three times and each time the food has been amazing.	
The diary that is supposed to be from the 18 th century might be real. It has passed a number of forensic and historical tests.	
The sum of two 2-digit numbers is always a 3-digit number.	

2. For each statement, write a counterexample that disproves the statement.
 - a. If a number is multiplied by itself, then the product is even.
 - b. Adding the same number to both the numerator and the denominator of a fraction does not change the fraction's value.



2.2.4 Wrap-Up: Cool Down

1. Write a statement that uses deductive reasoning.
2. Write a counterexample to the statement.

Unit 2, Lesson 3: Logical Conclusions



Benchmark

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can complete an argument that leads to a logical conclusion.
- ☐ I can determine if an argument leads to a logical conclusion.



2.3.1 Warm-Up: Bell Work

1. Students at Cypress Bay High School must have a 2.0 grade point average in order to participate in sports. Oliver has a 2.0 grade point average. Oliver can play lacrosse for the Cypress Bay Lighting.

Determine whether the scenario uses deductive or inductive reasoning.

2. Write a counterexample for the given statement.

If $\angle CRK$ is an acute angle, then $m\angle CRK = 84^\circ$.



2.3.2 Collaboration: Logical Arguments Using UNO

In the game of UNO, there are 112 cards. There are four sets of color cards: blue, green, red, and yellow, with digits 0 to 9. Each color has three types of specialty cards: Skip, Reverse, and Draw 2. There are two types of Wild cards: Wild and Wild Draw 4.

For each player's turn, the player matches the top card on the discard pile either by number, color, or word. If the player doesn't have anything that matches, they must pick a card from the draw pile. If the player draws a card that can be played, then the player plays that card.

1. Work with your partner to determine if the series of steps is possible in the game. Each statement represents a card played.

Statement	Reason
1. Green 8	1. Given
2. Blue 8	2. Matches number
3. Blue 3	3. Matches color
4. Wild	4. Changes color to yellow
5. Yellow Skip	5. Matches color

2. Add three additional statements and provide the reason.

Statement	Reason
6.	6.
7.	7.
8.	8.



3. Ask your partner to verify that your steps are valid.
4. The series of steps shown is not possible. Correct the error.

Statement	Reason
1. Red 4	1. Given
2. Red 9	2. Matches color
3. Yellow 9	3. Matches number
4. Green 4	4. Changes color to green
5. Red 4	5. Matches number
6. Red Reverse	6. Matches color

5. Ensure that you and your partner found the same error and that the corrections are both correct.

Yolanda made the following corrections.

Statement	Reason
1. Red 4	1. Given
2. Red 9	2. Matches color
3. Yellow 9	3. Matches number
4. Green 4 Blue 9	4. Changes color to green Matches number
5. Red 4 Blue Skip	5. Matches number color
6. Red Reverse Blue 5	6. Matches color

6. Explain why you or your partner's corrections may not match Yolanda's corrections.

7. Work with your partner to create a logical argument that leads to a Red Draw 2.

Statement	Reason
1. Green 5	1. Given
2.	2.
3. Yellow ☐	3.
4.	4.
5.	5.
6. Green 8	6.
7.	7.
8.	8.
9.	9.
10. Red Draw 2	10.

8. With your partner, use the template provided by your teacher to fill in your reasons for 5, 8, and 10. Then switch your paper with another pair. Work with your partner to fill in the missing statements and reasons on the other pair's paper.
9. Work as a group of four to make sure that the two completed templates correctly lead to the conclusion.



2.3.3 Activity: Taking Turns

Complete the UNO proofs. All UNO cards are available except for the Wild or Wild Draw 4 card. The only time a Wild card can be used is when it is already written in the proof.

Each proof must have at least three number changes and three color changes.



2.3.4 Wrap-Up: Reflection

1. How many of the UNO proofs did the class correctly complete?
2. Which steps in the UNO proof were the easiest to complete?
3. Which steps in the UNO proof were hardest to complete?



Unit 2, Lesson 4: Hypothesis and Conclusion



Benchmark

MA.912.LT.4.3 Identify and accurately interpret “if...then,” “if and only if,” “all” and “not” statements. Find the converse, inverse, and contrapositive of a statement.



Learning Targets

- ☐ I can identify the hypothesis of a statement.
- ☐ I can identify the conclusion of a statement.
- ☐ I can write the converse, inverse, and contrapositive of a conditional statement.



2.4.1 Warm-Up: Prototype Fabricator

Jeff Tiedeken is a prototype fabricator. He creates metal prototypes through bending, cutting, welding, and assembling.



Jeff states that most people who come in for help do not have an idea of what they need, so his first job is to understand the problem.

1. How does understanding the problem help?

Jeff said that once he understands the problem, he then solves the problem with other people.

2. Why do you think discussing a problem with others can be helpful?



2.4.2 Guided Instruction: Hypothesis and Conclusion

Often information is given as an if-then statement. A few examples are shown.

- If you leave the lights on, then it will take longer for people to fall asleep.
- If a goldfish is not exposed to light, then it will lose its color.
- If a quadrilateral has 4 right angles, then it is a rectangle.

1. Write a new if-then statement using a different phrase following "if."

If _____, then it
will take longer for people to fall asleep.

2. Write a new if-then statement using a different phrase following "then."

If a quadrilateral has 4 right angles, then _____
_____.

A **conditional statement** is a statement in the format of "If P , then Q ." The phrase P is the hypothesis. The phrase Q is the conclusion.

3. What is the hypothesis for “If a quadrilateral has 4 right angles, then it is a rectangle”?

4. Write a conclusion for each hypothesis.

a. If yesterday was Sunday, then _____
_____.

b. If you see lightning, then _____

_____.

5. William Camden wrote, “The early bird catches the worm.” Rewrite Camden’s statement as a conditional statement.



2.4.3 Exploration: Switching the Hypothesis and Conclusion

1. Draw a square or a triangle.

2. If you drew a square, then draw a circle inside the square.

If you drew a triangle, then draw a circle outside the triangle.

3. Discuss with your partner what happens when the hypothesis and conclusion of each conditional are switched.

- If you drew a circle inside the square, then draw a square.
- If you drew a circle outside the triangle, then draw a triangle.

Summarize your discussion.



2.4.4 Guided Instruction: Converse, Inverse, and Contrapositive

The **converse statement** is a statement in which the given conditional, "If P , then Q ," changes to "If Q , then P ."

1. Write the converse of "If it is raining, then the grass is wet."
2. Write a counterexample that shows the converse is not true.

The **inverse statement** is a statement in which the given conditional, "If P , then Q ," changes to "If not P , then not Q ."

3. Complete the inverse for "If it is raining, then the grass is wet."
If it is _____ raining, then the grass is _____ wet.
4. Write a counterexample that shows the inverse is not true.

The **contrapositive statement** is a statement in which the given conditional, "If P , then Q ," changes to "If not Q , then not P ."

5. Write the contrapositive for "If it is raining, then the grass is wet."



2.4.5 Guided Practice: Converse, Inverse, and Contrapositive

Write the converse, inverse, and contrapositive for each conditional.

1. "If you're not making mistakes, then you're not doing anything." (Quote by John Wooden, an American basketball player and coach)

Converse	
Inverse	
Contrapositive	

2. "If you think you're always in control, then you're not going fast enough." (Quote by Mario Andretti)

Converse	
Inverse	
Contrapositive	



2.4.6 Wrap-Up: Cool Down

1. Rewrite the following quote by Michelle Obama as a conditional statement.

"When someone is cruel or acts like a bully, you don't stoop to their level."

2. Complete the table for the conditional statement shown.

If you have ridden the roller coaster SheiKra, then you have been to Busch Gardens.

Hypothesis	
Conclusion	

3. Write the converse, inverse, and contrapositive for the conditional statement shown.

If a fruit is yellow, then it is a banana.

Converse	
Inverse	
Contrapositive	

Unit 2, Lesson 5: Conditionals and Biconditionals



Benchmark

MA.912.LT.4.3 Identify and accurately interpret “if...then,” “if and only if,” “all” and “not” statements. Find the converse, inverse, and contrapositive of a statement.



Learning Targets

- ☐ I can write the converse, inverse, and contrapositive of a mathematical statement.
- ☐ I can determine if a statement is biconditional.
- ☐ I can write a definition as a biconditional.



2.5.1 Warm-Up: Streaming Payouts

The payouts to a musician for each time a song is streamed on multiple streaming platforms in 2019 is shown.

Platform	Payout Per Stream
Napster	\$0.01900
Tidal	\$0.01250
Apple Music	\$0.00735
Google Play	\$0.00676
Deezer	\$0.00640
Spotify	\$0.00437
Amazon	\$0.00402
Pandora	\$0.00133
YouTube	\$0.00069

1. Determine the payout for a musician whose song was streamed 100,000 times on each platform.

Platform	Payout for 100,000 Streams (in dollars)
Napster	
Tidal	
Apple Music	
Google Play	
Deezer	
Spotify	
Amazon	
Pandora	
YouTube	

2. How many streams on Spotify did it take to for a musician to make \$1 in 2019?



2.5.2 Activity: True or False

For each conditional statement given, consider whether its converse, inverse, and contrapositive statements are true or false. Record the results in the table for each conditional.

- Conditional:** If all of the angles of a triangle are less than 90° , then the triangle is an acute triangle.

Relationship to Conditional	True or False
Converse	
Inverse	
Contrapositive	

- Conditional:** If the sum of two angles is 90° , then the angles are complementary.

Relationship to Conditional	True or False
Converse	
Inverse	
Contrapositive	

- Conditional:** If a polygon has six sides, then it is a hexagon.

Relationship to Conditional	True or False
Converse	
Inverse	
Contrapositive	

- Conditional:** If a line or ray divides an angle into two equal angles, then the line or ray is the bisector of the angle.

Relationship to Conditional	True or False
Converse	
Inverse	
Contrapositive	

- Conditional:** If lines that lie on the same plane do not intersect, then the lines are parallel.

Relationship to Conditional	True or False
Converse	
Inverse	
Contrapositive	

- Conditional:** If the opposite sides of a quadrilateral are parallel, then the quadrilateral is a parallelogram.

Relationship to Conditional	True or False
Converse	
Inverse	
Contrapositive	

- Conditional:** If $\overrightarrow{XY}$ is in the interior of $\angle WXZ$, then $m\angle WXY + m\angle YXZ = m\angle WXZ$.

Relationship to Conditional	True or False
Converse	
Inverse	
Contrapositive	



8. **Conditional:** If two planes intersect, then their intersection is exactly one line.

Relationship to Conditional	True or False
Converse	
Inverse	
Contrapositive	

9. **Conditional:** If two lines have the same slope, then they are parallel.

Relationship to Conditional	True or False
Converse	
Inverse	
Contrapositive	



2.5.3 Bring It Together: Biconditionals

1. How many of the conditionals from the previous Activity were considered true for all of the statements — converse, inverse, and contrapositive?
2. For how many conditionals was the converse of the conditional considered true?

A conditional statement is considered **biconditional** when both the statement and its converse are true. In other words, if both directions of the statement are true – “If P , then Q ” and “If Q , then P ” – the statement is biconditional.



Biconditional statements can be written in the format of “**if and only if**.”



A **definition** is an exact statement, written formally, of the meaning or description of a word. Even though definitions in mathematics are often written as conditional statements, all definitions are actually biconditional statements.

3. Write each conditional using “if and only if.”
 - a. If the sum of two angles is 90° , then the angles are complementary.
 - b. If the opposite sides of a quadrilateral are parallel, then the quadrilateral is a parallelogram.



2.5.4 Your Turn!

1. Complete the table for the conditional statement.

If all four angles at the point of intersection of two lines are right angles, then the lines are perpendicular.

Relationship to Conditional	Statement	True or False
Converse		
Inverse		
Contrapositive		

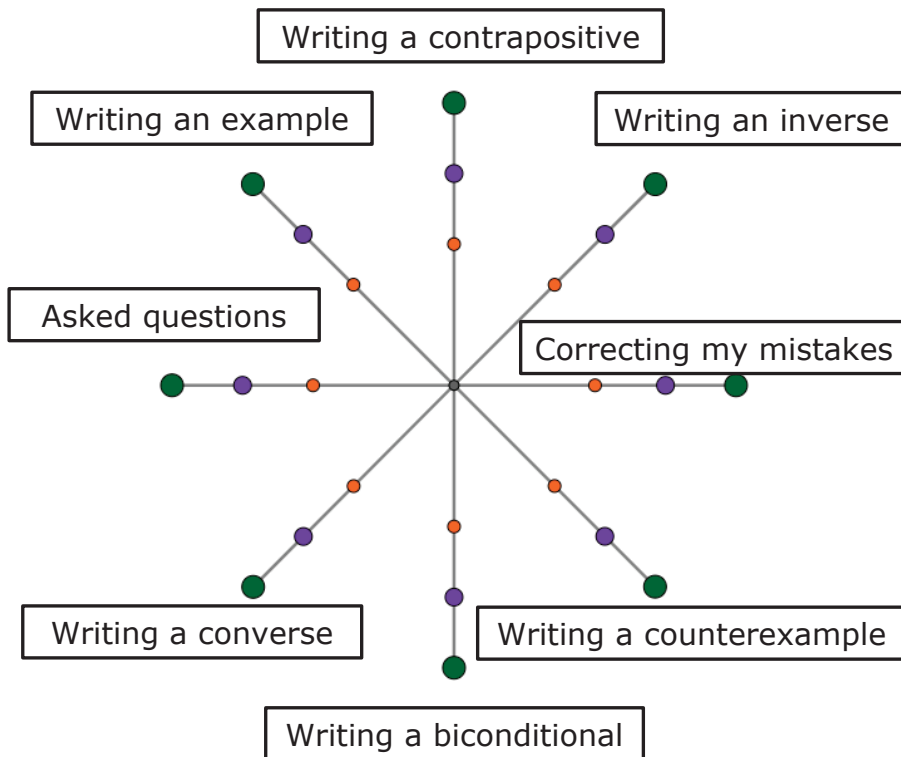
2. Explain why the conditional statement is biconditional.
3. Write the conditional using “if and only if.”



2.5.5 Wrap-Up: Reflection

- Place an **X** on the circle that shows what you learned or completed during today's lesson.

	I am confident I know how to do this!	OR	I always did this!
	I am fairly sure I know how to do this.	OR	I tried this.
	I need support on this.	OR	I had trouble with this.



Unit 2, Lesson 6: "All" and "Not" Statements



Benchmark

MA.912.LT.4.3 Identify and accurately interpret "if...then," "if and only if," "all" and "not" statements. Find the converse, inverse, and contrapositive of a statement.



Learning Targets

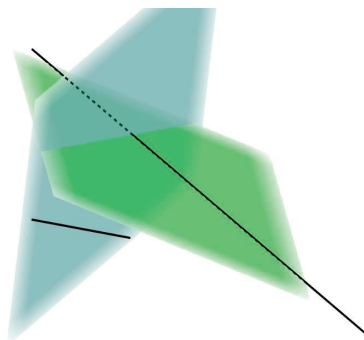
- ☐ I can write and interpret an "all" statement.
- ☐ I can write and interpret a negation.



2.6.1 Warm-Up: Bell Work

1. Complete the table for the conditional statement.

If two lines are skew, then the lines do not lie on the same plane.



Relationship to Conditional	Statement	True or False
Converse		
Inverse		
Contrapositive		

2. If the conditional statement is biconditional, then write it as a biconditional. If the conditional statement is not biconditional, then explain why not.



2.6.2 Guided Instruction: "All" and "Not" Statements

Mathematical statements may not be conditional or biconditional. Mathematical statements could be written using "all." The statement must be universally true to use the word "all."

1. Discuss with your partner if the statement, "All quadrilaterals have four sides" is universally true. Summarize your discussion.
2. Rewrite "All quadrilaterals have four sides" as an if-then statement.
3. Discuss with your partner why the statement "Some quadrilaterals have four sides" is not true. Summarize your discussion.

4. Determine if each statement is true.

Statement	True or False
All dogs are Labradors.	
All polygons have congruent sides.	
All cats are mammals.	

5. Rewrite each false statement using “some.”

6. Write each statement as an if-then statement.

Statement	If-Then
All dogs are Labradors.	
All polygons have congruent sides.	
All cats are mammals.	

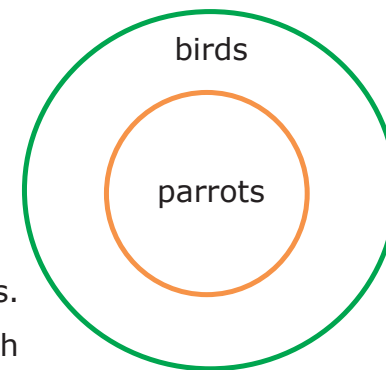
It may be helpful to understand “all” statements using Venn diagrams.



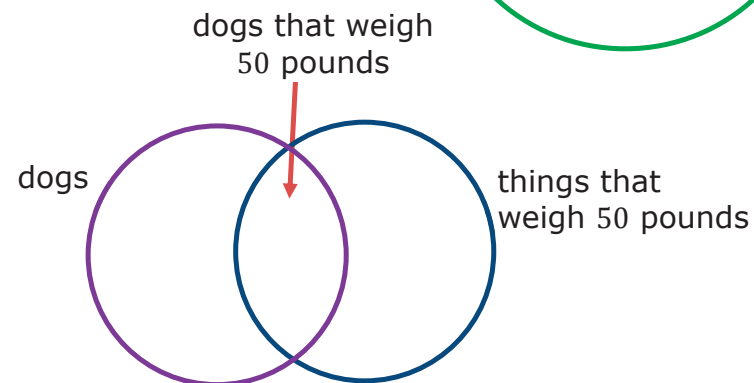
A **Venn diagram** is a diagram representing mathematical or logical sets pictorially as circles. Common elements of the sets are within the overlap of the circles.

7. The Venn diagram and a statement are shown.

All parrots are birds.

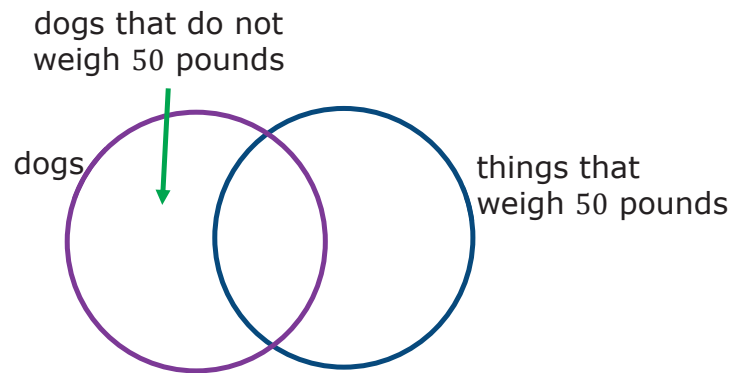


Some dogs weigh 50 pounds.



Discuss with your partner how the Venn diagrams help to show the difference between “all” and “some.” Summarize your discussion.

8. The Venn diagram below shows where dogs that do not weigh 50 pounds would be placed.



Write a statement that matches the Venn diagram.



The **negation** of a statement is the opposite of the statement. If the statement is true, then the negation must be false.

9. Use the definition of negation to write the negation of each statement.
- Today is Wednesday.
 - Seo-jun is not wearing blue.

10. Rolando and Rhianna both wrote the negation of the statement, " $\sqrt{5}$ is an irrational number."

Rolando	Rhianna
$\sqrt{5}$ is not an irrational number.	$\sqrt{5}$ is a rational number.

- Discuss with your partner both negations. Write your observations and questions for the class.
- Based on the class discussion, which student has written a correct negation?



2.6.3 Guided Practice: "All" Statements and Negation

1. Determine if each statement is true.

Statement	True or False
All congruent angles have the same measure.	
All odd numbers are prime.	
All points that lie in a plane are collinear.	

2. Rewrite each false statement using "some."

3. Write each statement as an if-then statement.

Statement	If-Then
All congruent angles have the same measure.	
All odd numbers are prime.	
All points that lie on a plane are collinear.	

4. Write the negation of each statement.

Statement	Negation
$\triangle DTM$ is a right triangle.	
$m\angle TYW + m\angle WYK = 136^\circ$	
$\angle RWF$ and $\angle HYV$ are complementary angles.	



2.6.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept below.



I need help.



I got it.



I could help someone.

I can write the converse of a mathematical statement.			
I can write the inverse of a mathematical statement.			
I can write the contrapositive of a mathematical statement.			
I can determine if a statement is biconditional.			
I can write and interpret an "all" statement.			
I can write and interpret a negation.			



Unit 2, Lesson 7: Proofs – Part 1



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

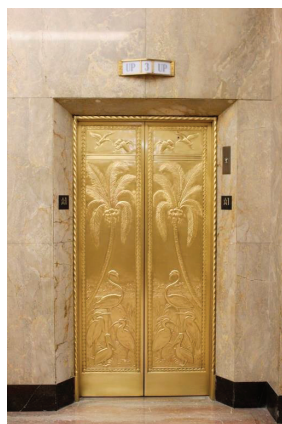
- ☐ I can relate deductive reasoning to completing proofs based on assumptions and given statements.
- ☐ I can recognize paragraph, flow-chart, and two column proofs.



2.7.1 Warm-Up: Miami Art Deco

A style known as Art Deco began in Paris in the mid-1920s. Art Deco buildings are characterized by their geometric shapes and symmetry. Two of the most famous examples of Art Deco architecture are the Empire State Building and Chrysler Building in New York City.

Photos of three locations in Miami that use this style of architectural design are shown.



1. Draw a line of symmetry on each picture.
2. Which depiction of Art Deco do you find most interesting?



2.7.2 Exploration: Proofs

A mathematical proof is an argument that shows something is true. A proof should begin with an assumption or a given statement.

1. A statement is given.

If two angles are supplementary to the same angle, then they are congruent to each other.

- a. What is the hypothesis?
- b. What is the conclusion?
- c. Draw a picture for the if-then statement.

The hypothesis is the given, or what is known to be true.

The conclusion will need to be shown, or proven, using deductive reasoning.

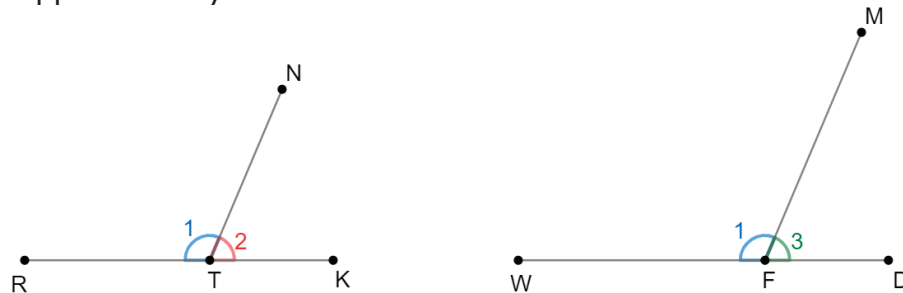
- d. Discuss with your partner why inductive reasoning is not appropriate for this context. Summarize your discussion.

One way to show the deductive reasoning that's necessary to prove the conclusion is a paragraph proof.



A **paragraph proof** is a narrative that gives the steps with justifications that prove a statement as true.

In the diagram, $\angle 1$ is supplementary to $\angle 2$, and $\angle 1$ is supplementary to $\angle 3$.



2. Work with your partner to complete the paragraph proof.

It is given that $\angle 1$ is supplementary to $\angle 2$, and $\angle 1$ is supplementary to $\angle 3$. By definition of _____ angles $m\angle 1 + m\angle 2 = 180^\circ$ and _____ $= 180^\circ$.

Since $m\angle 1 + m\angle 2 = 180^\circ$ and $180^\circ =$ _____, then $m\angle 1 + m\angle 2 =$ _____ by the transitive property.

The result of applying the subtraction property of equality is $m\angle 2 =$ _____. By definition of congruence, $\angle 2 \cong \angle 3$.

Another type of proof is the two-column proof.



A **two-column proof** consists of two columns of a chronological list of steps, called statements, and justifications, called reasons.

3. Complete the two-column proof.

Statement	Reason
1. $\angle 1$ is supplementary to $\angle 2$, $\angle 1$ is supplementary to $\angle 3$	1. Given
2. $m\angle 1 + m\angle 2 = 180^\circ$ _____ $= 180^\circ$	2. Definition of _____ angles
3. $m\angle 1 + m\angle 2 =$ _____	3. Transitive property
4. $m\angle 2 =$ _____	4. Subtraction property of equality
5. $\angle 2 \cong \angle 3$	5. Definition of congruence



If two figures have equal measurements, then by the **definition of congruence**, the figures are congruent. If congruence is shown through a series of transformations, then the **definition of congruence in terms of rigid motions** was applied.

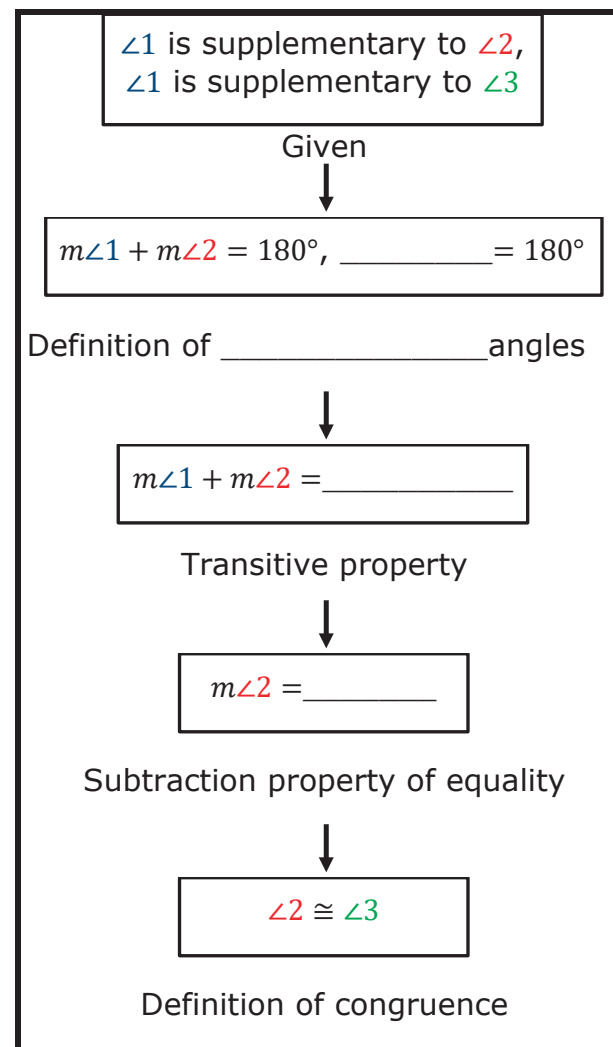
4. Discuss with your partner why the justification for step 5 is the definition of congruence. Summarize your discussion.

5. List similarities between the paragraph proof and the two-column proof.



A **flowchart proof** shows the structure of the proof using boxes and connecting arrows. The justifications are written beside or below the box.

6. Complete the flowchart proof.



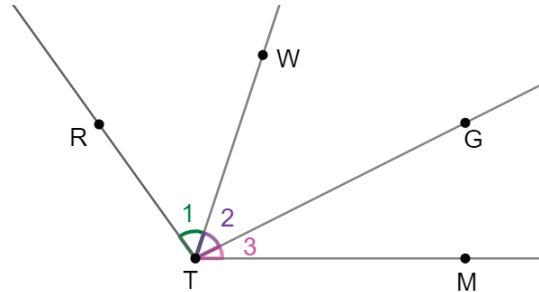
7. Which of the three types of proofs did you find the easiest to follow?



2.7.3 Guided Instruction: Flow Chart Proof

In the previous proof, each step depended on the step before it. But this is not always true.

1. Given: $m\angle 1 = m\angle 3$
 Prove:
 $\angle RTG \cong \angle WTM$



$$m\angle 1 = m\angle 3$$

$$m\angle 1 + m\angle 2 = m\angle RTG$$

$$m\angle 2 + m\angle 3 = m\angle WTM$$

$$m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$$

$$m\angle RTG = m\angle WTM$$

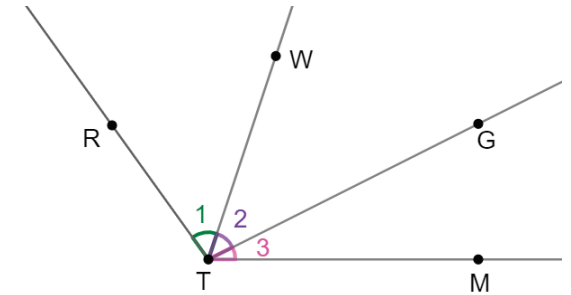
$$\angle RTG \cong \angle WTM$$



2.7.4 Guided Practice: Two-Column Proof

1. Given: $m\angle 1 = m\angle 3$
 Prove:
 $\angle RTG \cong \angle WTM$

Complete the two-column proof.

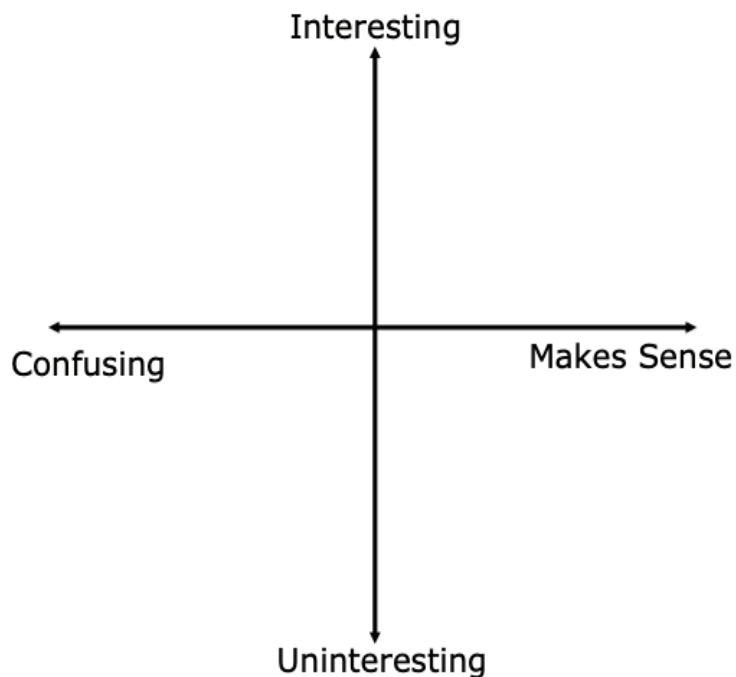


Statement	Reason
1. $m\angle 1 = m\angle 3$	1. Given
2. $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$	2.
3. $m\angle 1 + m\angle 2 = m\angle RTG$	3.
4. $m\angle 2 + m\angle 3 = m\angle WTM$	4.
5. $m\angle RTG = m\angle WTM$	5.
6. $\angle RTG \cong \angle WTM$	6.



2.7.5 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate how you feel about today's lesson.



Unit 2, Lesson 8: Proofs – Part 2



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

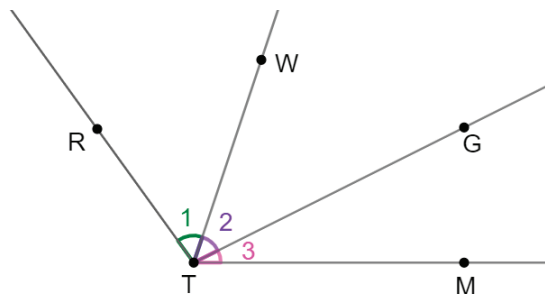
- ☐ I can reflect on the mathematical thinking found in a proof.
- ☐ I can reason about proofs using deductive reasoning, definitions, and postulates.



2.8.1 Warm-Up: Bell Work

Given: $m\angle 1 = m\angle 3$

Prove: $\angle RTG \cong \angle WTM$



1. Use the flowchart proof and the two-column proof from the previous lesson to write a paragraph proof.



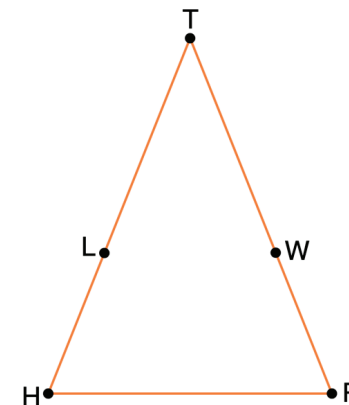
2.8.2 Collaboration: Brainstorming Before Writing A Proof

Work with your partner to complete the following.

Gabrielle and Samuel were given the following information.

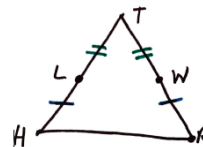
$\triangle THR$ is shown where $\overline{HL} \cong \overline{WR}$,
 $\overline{LT} \cong \overline{WT}$.

Prove that $\overline{HT} \cong \overline{TR}$.



Before Gabrielle and Samuel wrote their paragraph proof, they brainstormed. Their notes are shown below.

Gabrielle

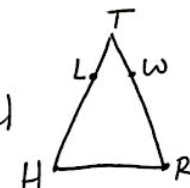


Given: $\overline{HL} \cong \overline{WR}$ $\overline{LT} \cong \overline{WT}$ Prove: $\overline{HT} \cong \overline{TR}$

$HT = HL + LT$ adding two segments to get the total
What is that called?

If using $=$ then it has to be changed to congruence.

Samuel



Given: $\overline{HL} \cong \overline{WR}$ $\overline{LT} \cong \overline{WT}$
Prove: $\overline{HT} \cong \overline{TR}$ don't worry, it's the same as $\overline{TR}$

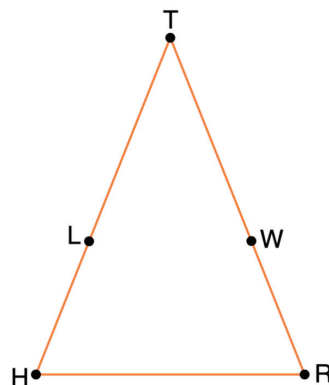
Basically if you add the two segments of each side you get the entire side.

1. Why did Samuel write "Don't worry it is the same thing as $\overline{RT}$?"
2. Gabrielle made some markings on the triangle she drew. What are her markings meant to show?
3. Gabrielle wrote " $HT = HL + LT$," but she could not remember what that was called. Samuel wrote "Basically if you add the two segments of each side you get the entire side." What postulate justifies Gabrielle's statements?

If you and your partner cannot remember the name of the postulate Gabrielle and Samuel are referring to, then look through your workbook to find it.

4. Gabrielle only wrote " $HT = HL + LT$," but Samuel mentioned "each side." Write the equation for the other side of the triangle.
5. Gabrielle wrote "If using $=$ then it has to be changed to congruence." What definition justifies changing from equality to congruence?
6. Apply the brainstorming of Gabrielle and Samuel to finish the paragraph proof. The first sentence has been written for you.

Given: $\overline{HL} \cong \overline{WR}$, $\overline{LT} \cong \overline{WT}$
Prove: $\overline{HT} \cong \overline{TR}$



It is given that $\overline{HL} \cong \overline{WR}$ and $\overline{LT} \cong \overline{WT}$.

7. Describe the most difficult part of writing the paragraph proof.

8. Describe how Gabrielle and Samuel's brainstorming was helpful.

9. Check your proof with another group and if necessary, resolve any differences.



2.8.3 Guided Instruction: Flowchart Proof

$\overline{AX}$ and $\overline{DY}$ are shown.

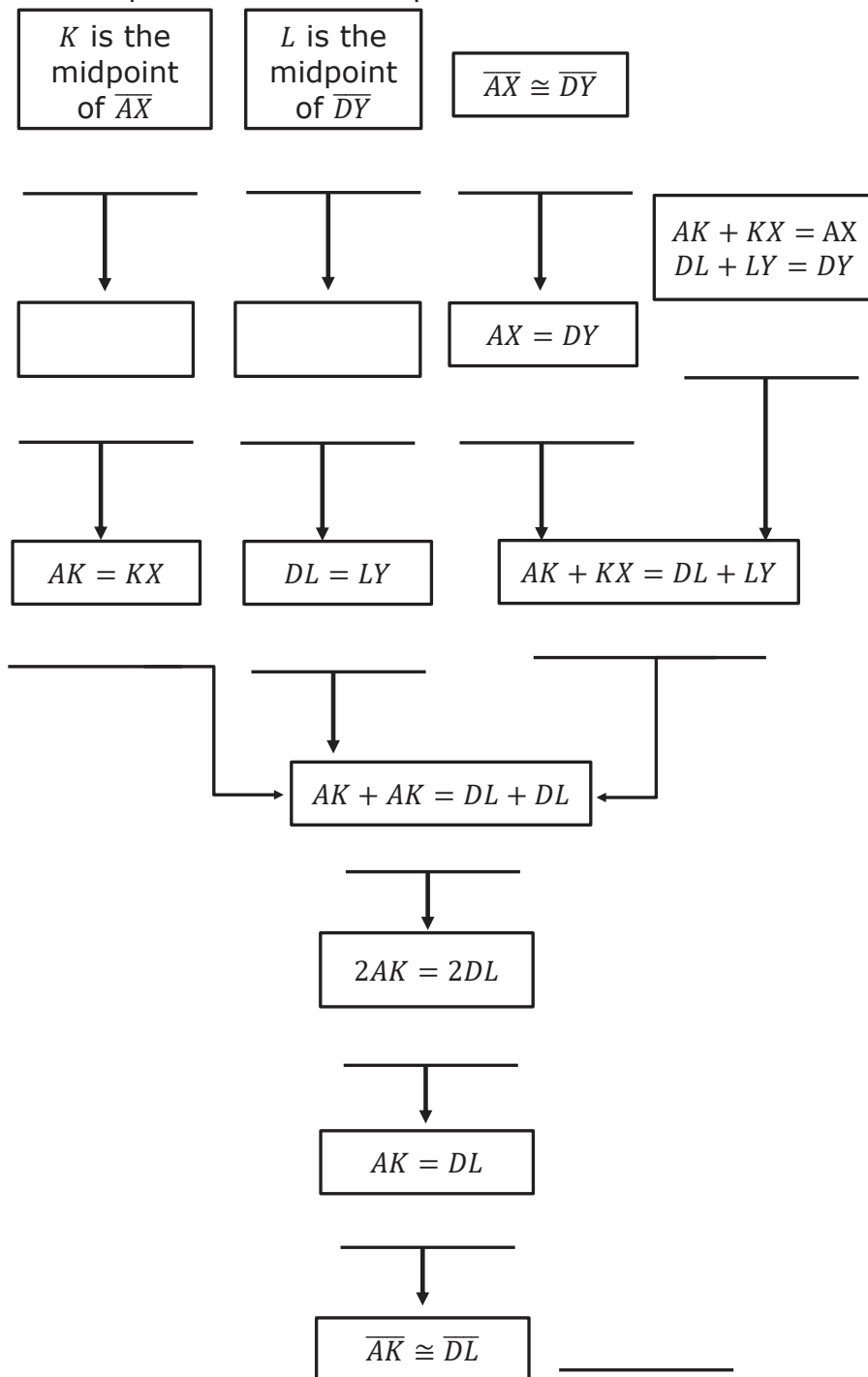


Given: K is the midpoint of $\overline{AX}$, L is the midpoint of $\overline{DY}$, and $\overline{AX} \cong \overline{DY}$

Prove: $\overline{AK} \cong \overline{DL}$

1. Brainstorm with your partner about how to write the proof. Then, write your brainstormed ideas in the space provided. Include any possible definitions or postulates that may need to be used to justify the steps.

2. Complete the flowchart proof.



2.8.4 Your Turn!

1. Use the previous flowchart to complete the two-column proof.

Statement	Reason
1. K is the midpoint of $\overline{AX}$ and L is the midpoint of $\overline{DY}$	1. Given
2.	2.
3. $AK = KX$ and $DL = LY$	3.
4. $\overline{AX} \cong \overline{DY}$	4. Given
5.	5. Definition of Congruence
6. $AK + KX = AX$ and $DL + LY = DY$	6.
7. $AK + KX = DL + LY$	7. Substitution property of equality
8. $AK + AK = DL + DL$	8.
9. $2AK = 2DL$	9.
10. $AK = DL$	10.
11. $\overline{AK} \cong \overline{DL}$	11.



2.8.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

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